

(12) **United States Patent**
Masson et al.

(10) **Patent No.:** **US 12,410,718 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **METHOD FOR PREPARING A ROOT OF A TURBOMACHINE BLADE**

(71) Applicant: **SAFRAN AIRCRAFT ENGINES**,
Paris (FR)

(72) Inventors: **Antoine Hubert Marie Jean Masson**,
Moissy Cramayel (FR); **Geneviève Mougey**,
Moissy Cramayel (FR)

(73) Assignee: **SAFRAN AIRCRAFT ENGINES**,
Paris (FR)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/694,821**

(22) PCT Filed: **Aug. 26, 2022**

(86) PCT No.: **PCT/FR2022/051613**
§ 371 (c)(1),
(2) Date: **Mar. 22, 2024**

(87) PCT Pub. No.: **WO2023/047030**
PCT Pub. Date: **Mar. 30, 2023**

(65) **Prior Publication Data**
US 2024/0287910 A1 Aug. 29, 2024

(30) **Foreign Application Priority Data**
Sep. 24, 2021 (FR) 2110098

(51) **Int. Cl.**
F01D 5/28 (2006.01)
F01D 5/00 (2006.01)
F01D 5/30 (2006.01)

(52) **U.S. Cl.**
CPC **F01D 5/282** (2013.01); **F01D 5/005**
(2013.01); **F01D 5/30** (2013.01); **F05D**
2230/80 (2013.01); **F05D 2300/603** (2013.01)

(58) **Field of Classification Search**
CPC . F01D 5/30; F01D 5/282; F01D 5/005; F05D
2230/80
See application file for complete search history.

(56) **References Cited**
U.S. PATENT DOCUMENTS
2007/0048142 A1* 3/2007 Dambrine F01D 5/3092
416/219 R
2013/0302173 A1 11/2013 Booze et al.

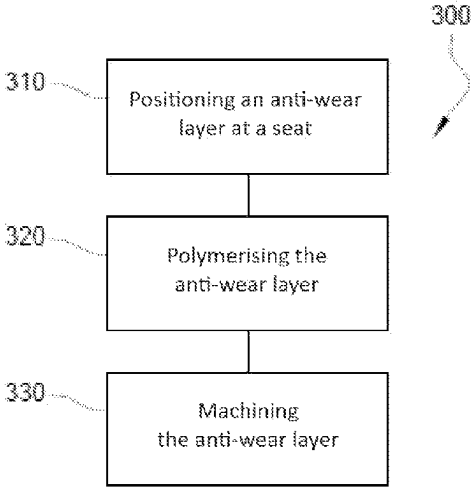
FOREIGN PATENT DOCUMENTS
EP 1 085 172 A2 3/2001
FR 2 890 126 A1 3/2007
(Continued)

OTHER PUBLICATIONS
Le Cloarec, FR 3049305 A1 English translation, 2017 (Year:
2017).*
(Continued)

Primary Examiner — Michael L Sehn
(74) *Attorney, Agent, or Firm* — Pillsbury Winthrop Shaw
Pittman LLP

(57) **ABSTRACT**
A method for preparing a root of a blade for mounting a
turbomachine blade, made of composite material, in a root
support, the method including positioning, on a side wall of
the root of the blade, an anti-wear layer including synthetic
fibres coated or impregnated with a bonding agent; polymer-
ising the bonding agent such that the anti-wear layer adheres
to the side wall of the root of the blade; and machining a free
outer surface of the anti-wear layer intended to be in contact
with the root support, so as to obtain a precision-ground
outer surface.

9 Claims, 2 Drawing Sheets



(56)

References Cited

FOREIGN PATENT DOCUMENTS

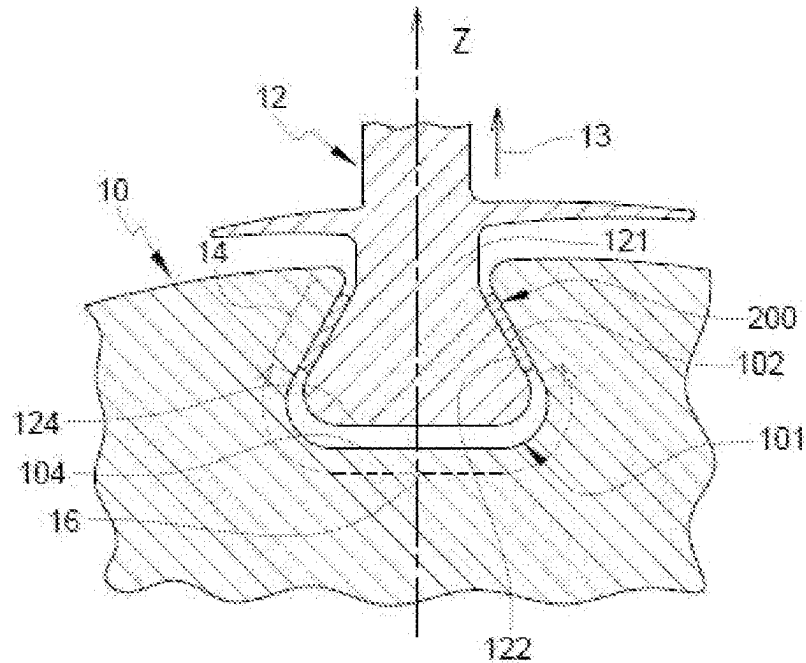
FR	2 890 684 A1	3/2007
FR	3 049 305 A1	9/2017
FR	3 059 323 A1	6/2018
WO	WO 2014/164859 A2	10/2014
WO	WO 2015/023345 A1	2/2015

OTHER PUBLICATIONS

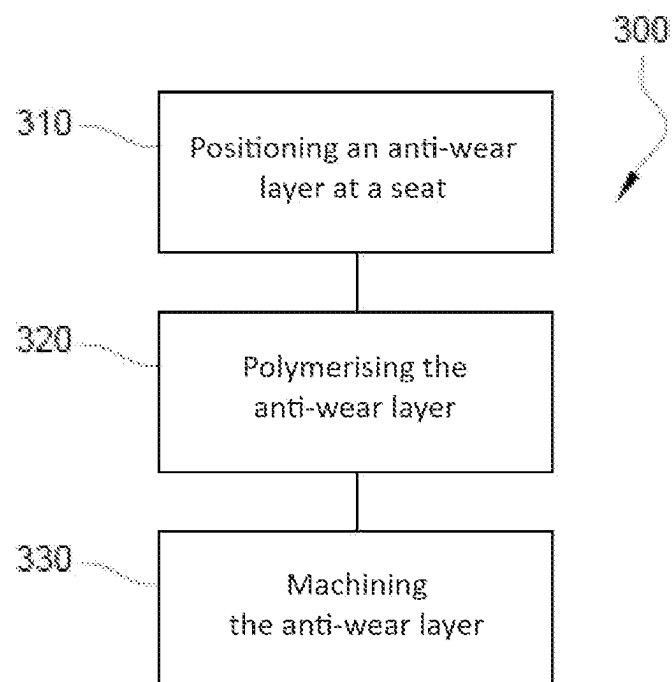
International Search Report as issued in International Patent Application No. PCT/FR2022/051613, dated Oct. 13, 2022.

* cited by examiner

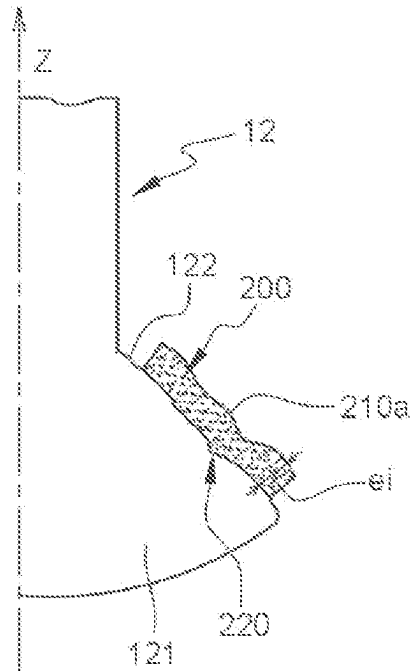
[Fig. 1]



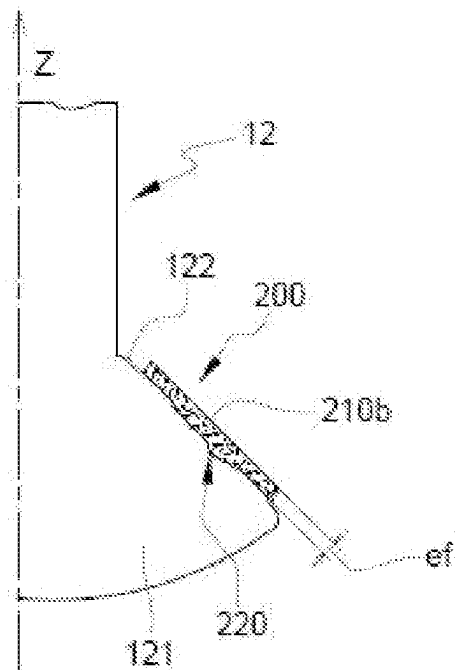
[Fig. 2]



[Fig. 3]



[Fig. 4]



METHOD FOR PREPARING A ROOT OF A TURBOMACHINE BLADE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is the U.S. National Stage of PCT/FR2022/051613, filed Aug. 26, 2022, which in turn claims priority to French patent application number 2110098 filed Sep. 24, 2021. The content of these applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD OF THE INVENTION

The technical field of the invention is that of the preparation of a turbomachine vane root for the purpose of mounting it in a root support.

The invention more particularly relates to preparing and mounting a fan vane in a fan disc of a turbomachine.

The invention also relates to a turbomachine vane obtained by such a method, as well as to a rotor assembly of a turbomachine.

TECHNICAL BACKGROUND

In the present application, the terms “upstream” and “downstream” are defined in relation to the normal direction of gas flow (from upstream to downstream) through a turbomachine.

The axis of rotation of a turbomachine rotor is also referred to as the “turbomachine axis” or “engine axis”. The axial direction corresponds to the direction of the axis of the turbomachine and a radial direction is a direction perpendicular to the axis of the turbomachine and intersecting this axis. Similarly, an axial plane is a plane containing the axis of the turbomachine, and a radial plane is a plane perpendicular to that axis.

Unless otherwise specified, the adjectives “internal”, “inner”, “external” and “outer” are used in the present application with reference to a radial direction so that the internal part of an element is, along a radial direction, closer to the axis of the turbomachine than the external part of the same element.

Conventionally, a turbomachine includes, from upstream to downstream, i.e. in the flow direction of the gas streams, a fan, one or more compressors, a combustion chamber, one or more turbines, and a nozzle for ejecting combustion gases leaving the turbine or turbines.

In a rotor system (i.e. an assembly integral with the rotor), the (moving) vanes are attached to a rotor disc by fastening systems, which may be straight or curved pinned fasteners, hammer fasteners or fir fasteners. These fastening systems can be described as devices in which the vane roots form male parts of the system and are held radially in female parts of the system, provided at the external periphery of the disc and commonly referred to as cells.

Upon rotating the rotor, the vanes are mainly subjected to centrifugal forces as well as axial aerodynamic forces and the vane roots come to abut against the parts of the disc bordering the external opening of the cells, under the effect of the centrifugal forces. The surfaces of the vane roots and the disc, abutting against each other, are commonly referred to as “seats”. These seats are subject to pressure (resulting from said forces, applied to the surface of these seats). It can be estimated that this pressure depends, to a first approximation, on the square of the speed of rotation of the rotor.

It is therefore understood that the variations in the speed of rotation of the rotor during the operating cycle of the turbomachine: from standstill to full throttle, passing through the particular intermediate speeds, induce variations in pressure at the seats previously defined. These variations in pressure, combined with the elastic deformation of pieces in contact with each other, cause relative movements between the vane root and the disc. These relative movements, called sliding or separation depending on their nature, induce wear phenomena on the vane or disc seats when they are repeated. The dynamic movements of the vanes at a given rotation speed (responses of the vanes to alternating loads of a harmonic or transient nature) can also be attributed a contribution to the wear phenomenon of said seats.

Yet these wear phenomena are detrimental to the life of the turbomachine. So-called “anti-wear” solutions, i.e. delaying the onset of wear at the contact interfaces, can be adopted, including solutions based on the introduction of a third body, called a foil, between the vane roots and the disc. This foil makes it possible, especially, to double the contact interfaces (from one vane/disc interface to two vane/foil and foil/disc interfaces) and to reduce the relative movements between the pieces in contact with each other, thereby reducing wear during operation.

One example of a known foil of the above-mentioned type is described in document FR 2890684. This foil is made entirely of metal and is a suitably folded metal sheet.

One alternative to the foil, described in document FR2890126, consists in introducing as a third body a non-metal “anti-wear” film including resistant fibres impregnated with resin at the vane/disc contact interface.

Some of the vanes are made of composite material, but their surface condition is not sufficiently flat and it is not possible (or very difficult) to machine these seats without degrading and/or altering the fibres. The coating that protects the composite from wear is expensive, its range of application often generates non-compliances (in location) and does not completely solve the problem of surface condition. This leads to extra costs (factory repairs, etc.).

These solutions are therefore not entirely satisfactory.

SUMMARY OF THE INVENTION

One purpose of the invention is to provide one alternative solution to the solutions of the state of the art, which is more effective than the solutions previously described, in terms of “anti-wear” performance, so as to provide better protection for the vane and disc seats and make it possible to guarantee satisfactory flatness.

Thus, the invention seeks to provide a method for preparing a vane root for mounting a turbomachine vane made of composite material in a root support, for example a rotor disc cell, in order to improve mechanical strength of the pieces at contact pressure, while allowing very easy and rapid implementation of this method.

To this end, one object of the invention is a method for preparing a vane root for mounting a turbomachine vane made of composite material in a root support, characterised in that it comprises:

- a first step of positioning on a side wall of the vane root an anti-wear layer including synthetic fibres coated or impregnated with an adhesive agent;
- a second step of polymerising the adhesive agent so that said anti-wear layer adheres to the side wall of the vane root;

3

a third step of machining a free external surface of said anti-wear layer intended to be in contact with the root support, so as to obtain a ground external surface.

Advantageously, the machining step makes it possible to decrease thickness of the anti-wear layer so as to shift from an initial thickness (e_i) to a final thickness (e_f), the initial thickness corresponding to the thickness of the anti-wear layer during the positioning step.

Advantageously, the initial thickness of the anti-wear layer (**200**) is greater than 1 millimetre. Thus, the anti-wear layer is sufficiently thick to be handled easily and has sufficient material to be ground so as to obtain a final thickness in the order of a few tenths of one millimetre.

Advantageously, the machining step is carried out by taking the machining reference frame of the vane root as the machining reference frame of the free external surface of said anti-wear layer.

Advantageously, the anti-wear layer includes at least one woven ply of synthetic fibres, for example glass fibres.

Advantageously, before positioning the anti-wear layer, the synthetic fibres thereof are pre-coated with a surface or bulk adhesive agent.

Preferably, the adhesive agent is a thermo-polymerisable adhesive.

Advantageously, the step of polymerising the adhesive agent is carried out by placing the assembly comprising at least the vane root and the anti-wear layer in an autoclave heated to a temperature allowing polymerisation of the adhesive agent, for a predetermined minimum period of time.

According to one alternative embodiment of the method according to the invention, the vane root can be coated with a metal foil. In this case, the anti-wear layer is positioned on a surface of the foil intended to be in contact with the root support.

Another object of invention is a method for mounting the vane root of a turbomachine in a root support including a preparation step according to the method of the invention.

Another object of invention is a method for repairing a turbomachine vane root consisting in partially or totally removing a damaged anti-wear member and preparing the vane root according to the method of the invention.

Thus, the preparation method according to the invention is both integratable into a method for mounting a turbomachine vane root as well as into a vane root seat repair method.

Another object of the invention is a turbomachine vane having an anti-wear member with a ground surface for being more effective than solutions in the state of the art in terms of "anti-wear" performance and flatness, so as to provide better protection for the vane and disc seats.

This purpose is achieved by the fact that the root of the vane includes, at a seat, at least one anti-wear layer including synthetic fibres and having a ground external surface. Thus, the surface condition, and especially the flatness, of the external surface intended to come into contact with the seat of the root support is improved and controlled.

Indeed, improving the surface condition, and especially the flatness of the seats, makes it possible to improve mechanical strength of the interfaces with respect to the aforementioned loads at the vane/disk interfaces.

Another object of the invention is a turbomachine rotor assembly including a rotor disc having cells on its external periphery and a plurality of vanes according to the invention attached through their roots in said cells.

The invention also relates to a turbomachine including a rotor assembly according to the invention.

4

The invention and its different applications will be better understood upon reading the following description and upon examining the accompanying figures.

BRIEF DESCRIPTION OF THE FIGURES

The figures are only set forth by way of indicating and in no way limiting purposes of the invention.

FIG. 1 represents a partial transverse cross-section view of a fan vane showing mounting of the vane root in a cell of a rotor disc, the vane being subjected to centrifugal forces.

FIG. 2 is a block diagram illustrating the main steps of the method for preparing the vane root according to the invention.

FIG. 3 is a schematic partial cross-sectional view of the root of a fan vane according to the first step of the method according to the invention.

FIG. 4 is a schematic partial cross-sectional view of the root of a fan vane according to the third step of the method according to the invention.

Unless otherwise specified, a same element appearing in different figures has a single reference.

DETAILED DESCRIPTION

FIG. 1 represents a partial cross-sectional view of a fan vane **12**, the root **121** of which is housed in a root support, and more particularly a cell **101** of a rotor disc **10**, the vane being subjected to centrifugal forces.

The invention will be mainly described by taking as an example mounting of a fan vane in a root support formed by a fan rotor disc.

However, the present invention is not limited to mounting of fan vanes in a fan rotor disc. Indeed, the invention is also applicable to other types of mounting of turbomachine movable vanes. Thus, the invention is applicable to any composite piece where flatness is important, in particular all composite parts interfacing with a metal piece for which grinding is necessary.

These composite pieces are formed from a woven preform which is injected with a resin. This preform can be a 2D or 3D preform. The fan vane is manufactured from a 3D woven preform.

Complementarily, the present invention applies to fixed vanes and supports associated therewith, such as casings or flanges supporting fixed vanes.

Conventionally, a fan movable vane **12** is formed by two assemblies: a lower part called the root **121**, and an upper part formed by the blade (not represented).

The fan movable vane **12** extends along a longitudinal axis Z. It should be noted that the longitudinal axis Z of the fan movable vane **12** is perpendicular to the axis of rotation of the fan.

Each cell **101** of the rotor disc **10** has a shape substantially complementary to the shape of the root **121** of the vane **12** to form a dovetail-type mounting.

The cells **101** are radially equidistantly distributed along the circumference of the rotor disc **10**, and the opening of the cell is directed outwardly.

As can be seen in [FIG. 1] showing an enlarged portion of the transverse cross-section of the rotor disc **10**, i.e. a cross-section orthogonal to the axis of rotation of the fan, each root **121** of a vane **12** has a symmetrical contour with two side walls **122**, forming the flanks of the root **121**, diverging from one another from the body of the vane **12** towards the free end of the root **121** of the vane **12**, to a bottom wall **124** substantially parallel to the axis of rotation

of the fan and to the periphery of the rotor disc **10**, and orthogonal to the main longitudinal direction **Z** of the corresponding vane **12**.

The cells **101** have a similar shape with side walls **102** outwardly tilted from the circumference towards the internal portion of the rotor disc **10** to a bottom wall **104**.

The dimensions of the root **121** of the vane **12** and of the cell **101** are such that when the rotor disc **10** is at rest, the root **121** is retained in the cell **101**, the bottom wall **124** of the root **121** then being able to touch the bottom wall **104** of the cell **101**. To avoid too great a degree of freedom in the radial movement of the blade, a shim may be present.

When the rotor disc **10** is operational during operation of the turbomachine, rotation of the rotor disc **10** about the central axis causes the vanes **12** to move radially outwardly as a result of centrifugal forces, i.e. in the direction of the arrow **13** in [FIG. 1], substantially parallel to the longitudinal axis **Z** of the vane **12**. At this point, the side walls **122** of the root **121** of the vane **12** bear against the side walls **102** of the cell **101**, which ensures that the vane **12** is retained inside the cell **101**, i.e. is connected to the rotor disc **10**.

Each time the speed of rotation of the rotor disc **10** is modified, the sliding movement of the root **121** of the vane **12** combined with the contact pressure of the root **121** and the coefficient of friction between the materials of the vane **12** and the disc **10**, generate shearing forces on both the rotor disc **10** and the vane **12**, one moving radially with respect to the other by 1 to a few millimetres.

In particular, as is apparent from [FIG. 1], there is in operation an interface zone called a "seat" subject to high forces, designated under the reference sign **14**, between the side wall **122** of the root **121** and the side wall **102** of the cell **101**, as well as a non-contact region indicated under the reference sign **16**, which is not subject to any mechanical contact stress during rotation of the rotor disc **10**.

In accordance with the invention, the fatigue and wear damage that occurs on the contacting surfaces of the side walls **102** and **122** as a result of the relative movement between the root **121** of the vane **12** and the cell **101** of the rotor **10** is reduced by the use of a ground anti-wear layer **200** positioned at the seats **14** between the side wall **122** of the root **121** and the side wall **102** of the cell **101**.

More precisely, the anti-wear layer **200** is comprised of synthetic fibres. For example, the anti-wear layer **200** is comprised of at least one woven fabric of synthetic fibres.

By way of example, the anti-wear layer **200** is comprised of at least one woven fabric made from glass fibres and/or aramid fibres.

This anti-wear layer **200** adheres to the external surface of the root **121** with an adhesive agent coating or impregnating the synthetic fibres of the layer **200**, at least on the face of the layer **200** that is to adhere.

The adhesive agent is, for example, a thermo-polymerisable synthetic adhesive.

The adhesive agent is for example a resin and advantageously a phenolic or polyurethane resin.

As is visible in [FIG. 3], the anti-wear layer **200** has a first external face **210a** and a second internal face **220** to be adhered to a side wall **122** of the root **121**.

This anti-wear layer **200** has a significant initial thickness e_i , greater than 1 millimetre, and advantageously in the order of 2 millimetres. The thickness e_i , referred to as the initial thickness, of the anti-wear layer **200** corresponds to its thickness upon manufacturing and upon adhering to the root **121** of the vane **12**.

During a first step **310** of the preparation method **300**, the main steps of which are illustrated in [FIG. 2], an anti-wear

layer **200** is positioned on each side wall of the root **121** at the seats **14**, as illustrated in [FIG. 3].

This anti-wear layer **200** adheres to the external surface of the root **121** via the adhesive agent coating or impregnating the synthetic fibres of the layer **200**.

According to one alternative embodiment, it is also possible to impregnate the synthetic fibres of the anti-wear layer **200** directly (i.e. the anti-wear layer is not pre-impregnated before being positioned), or complementarily, when the layer is in position on the root **121**. In this case, impregnation is carried out at the same time as injection of the vane preform.

A second step **320** consists in polymerising the adhesive agent. In this second step **320**, the assembly formed by the root **121** of the vane **12** and the anti-wear layer **200** is pressurised with the adhesive agent until the adhesive agent is polymerised.

The pressure used is in the order of 7 to 14 MPa. This ensures good adhesion of the adhesive over the entire surface as well as discharge of the solvents.

Polymerisation is carried out, for example, at a temperature of between 150° C. and 180° C. for a period of time of 1 hour.

A thick anti-wear layer **200** having thickness e_i is thus obtained, perfectly adhering to the side wall **122** of the root **121**.

At this stage of the preparation method, the thickness e_i of the anti-wear layer **200** is greater than the operating and mounting clearance between the root **121** and the cell **101**, so that mounting as it stands is not possible.

It may also be noticed that, following this polymerisation step schematically illustrated in [FIG. 3], the surface condition and especially flatness of the external face **210a** is substantially identical to the surface condition of the side wall **122** of the root **121**. At this stage, the surface condition of the anti-wear layer **200** has flatness defects.

After the polymerisation step **320**, the root **121** of the vane **12** can be cleaned to remove any excess adhesive agent that underwent creep, by sandblasting, for example with glass beads.

In a third step **330**, the external face **210a** of the anti-wear layer **200** adhered to the root **121** of the vane **12** is machined. Machining makes it possible to decrease thickness of the anti-wear layer **200** and to grind the external face **210a** so as to obtain a planar ground external face **210b** as illustrated in FIG. 4. This machining step makes it possible to control final thickness e_f of the anti-wear layer **200** and to improve surface condition as well as flatness tolerances of the external surface **210b**.

After machining, the anti-wear layer **200** has a final thickness e_f in the order of a few tenths of one millimetre, advantageously in the order of 0.3 millimetre.

Machining the external face **210a** of the anti-wear layer **200** is carried out using as a reference frame the machining reference frame of the side faces **122** of the root **121** of the vane **12**, which makes it possible to control orientation and thickness of the anti-wear layer **200**. This ensures better positioning of the root **121** of the vane **12** in the cell **101** of the rotor disc **10**, as well as better stress distribution.

Thus, by virtue of such a method for preparing the vane roots prior to mounting, it is possible to very significantly increase life time of the interfaces between the roots **121** of the vane **12** and the cells **101** of the rotor disc **10**, especially as a result of a better force distribution over the entire surface of the anti-wear layer **200**.

7

The preparation method **300** previously described can be integrated into a method for manufacturing a turbomachine vane.

The preparation method **300** previously described may also be integrated into a more overall method for mounting a vane **12** in a rotor disc **10** or even a rotor assembly.

The preparation method **300** previously described can also be integrated into a more overall method for repairing an anti-wear member at the junction between the vane root and the cell, regardless of whether initial mounting was carried out in accordance with the invention or with another type of anti-wear member, whether metallic or non-metallic.

The invention claimed is:

1. A method for preparing a root of a vane, the vane being a turbomachine vane made of composite material and to be mounted in a root support, the method comprising:

a first step of positioning on a side wall of the root of the vane an anti-wear layer including synthetic fibres coated or impregnated with an adhesive agent;

a second step of polymerising the adhesive agent so that said anti-wear layer adheres to the side wall of the root of the vane, and

after the second step, a third step of machining a free external surface of said anti-wear layer for being in contact with the root support, so as to obtain a ground external surface that forms at least a portion of the external surface of the root of the vane, the third step of machining decreasing a thickness of the anti-wear layer so as to shift from an initial thickness greater than 1 millimetre to a final thickness, the initial thickness of the anti-wear layer corresponding to the thickness of the anti-wear layer during the first step of positioning.

2. The method for preparing the root of the vane according to claim **1**, wherein the third step is carried out by taking a machining reference frame of the root of the vane as a machining reference frame of the free external surface of said anti-wear layer.

8

3. The method for preparing the root of the vane according to claim **1**, wherein the anti-wear layer includes at least one woven ply of synthetic fibres.

4. The method for preparing the root of the vane according to claim **1**, wherein the anti-wear layer includes at least one woven ply of glass fibres.

5. A method for repairing the root of the turbomachine vane made of composite material, comprising partially or totally removing a damaged anti-wear member and preparing the vane root according to the preparation method according to claim **1**.

6. The turbomachine vane made of composite material including the root prepared according to the preparation method according to claim **1**, wherein the root includes, at the side wall, the anti-wear layer including the synthetic fibres and the polymerised adhesive agent and having the ground external surface obtained by machining the free external surface of the anti-wear layer.

7. The method for preparing the root of the vane according to claim **1**, wherein the second step of polymerizing is performed at a temperature between 150° C. and 200° C. and under a pressure of 7 to 14 MPa for a minimum duration of 1 hour.

8. The method for preparing the root of the vane according to claim **1**, wherein the anti-wear layer comprises at least one woven ply of synthetic fibers selected from the group consisting of glass fibers, aramid fibers, and a combination thereof.

9. The method for preparing the root of the vane according to claim **1**, wherein the anti-wear layer is pre-impregnated with the adhesive agent before positioning on the vane root, and excess adhesive is removed after polymerization by a non-abrasive cleaning technique.

* * * * *